

Τμήμα Μηχανικών Ηλεκτρονικών Υπολογιστών & Πληροφορικής

Προχωρημένα Μαθηματικά Ι (ΕΕΙ_101)

Κατοίκον Εργασία: ΦΘΙΝΟΠΩΡΙΝΟ ΕΞΑΜΗΝΟ 2024-2025

Φοιτητής:

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Καθηγητής:

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1 Δεκεμβρίου, 2024

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Κατοικον Εφαρσια: Μαθηματικά: ΚΑΡΥΟΣ Γ.

1.1) 2)

β) $F(x)=2x^3-x$, $g(x)=\begin{cases}\frac{1}{x-1}, & x>1\\ 2x, & x\le 1\end{cases}$

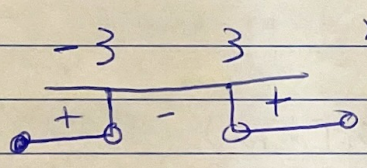
- $g(0)=0$ / $2\cdot 0=0$
- $g(\sqrt{3})=\frac{1}{\sqrt{3}-1}=\frac{1+\sqrt{3}}{2}$
- $g(7)=2\cdot 7=2$
- $g(-7)=2\cdot(-7)=0$
- $g(1/3)=2/3$
- $g(a)=\begin{cases}\frac{1}{a-1}, & a>1\\ 2a, & a\le 1\end{cases}$
- $g(\sqrt{a})=\begin{cases}\frac{1}{\sqrt{a}-1}, & a>1\\ 2\sqrt{a}, & a\le 1\end{cases}$
- $g(\frac{1}{a})=\begin{cases}\frac{1}{\frac{1}{a}-1}, & a>1\\ \frac{2}{a}, & 0<a<1\end{cases}$
- $g(a^2)=\begin{cases}2a^2, & a\le 1\\ \frac{1}{a^2-1}, & a>1\end{cases}$

Ασκ 3)

α) $F(x)=2x^3-x$, $x\in R$

β) $x^2-9\ge 0 \Rightarrow x^2\ge 9$

$x\ge\pm 3$



$AF=x\in(-\infty,3]\cup[3,+\infty)$

γ) $x\in R$, $x^2+9\ge 0$

δ) $2x-3\neq 0$

$x\neq 3/2 \Rightarrow AF=x\in R-\{3/2\}$

ε) $F(x)=\frac{2x^2-7}{2x-7}$, $2x-7\neq 0$

$2x\neq 7$

$x=\frac{7}{2}$

$AF=x\in R-\{\frac{7}{2}\}$

στ) $F(x)=\sqrt{\frac{x^2-4}{x-7}}$, $\frac{x^2-4}{x-7}\ge 0$, $x\neq 7$

$x=2$, $x=-2$

x		-2		1		2	
	-	0	+	‖	-	0	+

$AF=x\in[-2,7)\cup[2,+\infty)$

ζ) $F(x)=\sin x+\cos x$

$x\in R$

η) $F(x)=\cos^2\sqrt{x}$

$x\ge 0$, $x\in[0,+\infty)$

~~ε) $F(x)=\frac{2x^2-7}{2x-7}$ $2x\neq 7$ $x=\frac{7}{2}$ $AF=x\in R-\{\frac{7}{2}\}$~~

~~στ) $F(x)=\sqrt{\frac{x^2-4}{x-7}}$ $\frac{x^2-4}{x-7}\ge 0$~~

θ) $F(x)=\cos^3\sqrt{x}$

$x \geq 0$, $x \in [0, +\infty)$

ι) $f(x)=\sqrt{\sin x}$

$\sin x \geq 0$ $x \in [0,\pi] \cup [\pi, ?]$

ια) $F(x)=\frac{3}{2-\cos x}$ $2-\cos x \neq 0$

$\cos x \neq 2$ $x \in \mathbb{R}$

ιβ) $F(x)=\frac{3}{1-\cos x}$ $\cos x \neq 1$

$x \in \mathbb{R}$

4) a) ΠΤ: $y \in \mathbb{R}$

β) $y=\sqrt{x^2-9} \Rightarrow y^2=x^2-9 \Rightarrow x^2=y^2+9 \Rightarrow x=\sqrt{y^2+9}$

$y \in [0, +\infty)$

ε) $y=\frac{2x^2-7}{2x-7}$ $2x-7 \neq 0$ $x \neq 7/2$

$2xy-y=2x^2-7 \Rightarrow 2xy-2x^2=y-7$

$\Rightarrow x(2y-2x)=y-7$

$$x=\frac{y-7}{2y-2x}$$

$y \in \mathbb{R}-\{\}$

η)

1.2) 7γ) i) $(f+g)(x)=2x+\sqrt{x}$ $A_{fg}=[0,+\infty)$

ii) $(f-g)(x)=2x-\sqrt{x}$ $A_{fg}=[0,+\infty)$

iii) $(fg)(x)=2x\cdot\sqrt{x}$ $A_{fg}=[0,+\infty)$

iv) $(f/g)(x)=\frac{2x}{\sqrt{x}}$ $A_{fg}=(0,+\infty)$

v) $(g/f)(x)=\frac{\sqrt{x}}{2x}$ $A_{gf}=(0,+\infty)$

2γ) i)

$A_f \in \mathbb{R}$ $A_g \in [7,+\infty)$

$(f\circ g)(x)=f(g(x))=(\sqrt{x-7})^2$ $A_{fog}=\{x \in [7,+\infty) : \sqrt{x-7} \in \mathbb{R}$

ii) $(f\circ f)(x)=f(f(x))=x^{2^2}=x^4$ $A_{fof}=\{x \in \mathbb{R} : x^2 \in \mathbb{R}\}=\mathbb{R}$

iii) $(g\circ f)(x)=g(f(x))=\sqrt{x^2-7}$ $A_{gof}=\{x \in \mathbb{R} : x^2 \in [7,+\infty)\}=\mathbb{R}-[7,+\infty)$

iv) $(g\circ g)(x)=g(g(x))=\sqrt{\sqrt{x-7}-7}$ $A_{gog}=\{x \in [7,+\infty) : \sqrt{x-7} \in [7,+\infty)\}$

$x \geq 7$ $\sqrt{x-7} \geq 0$ $x-7 \geq 7$ $x \geq 14$

3α) i) α) $fogoh \Rightarrow fog = f(g(x)) = \frac{1}{x^2-1}$

$foh(h) = \frac{1}{x^2-1}$

$\Rightarrow Dfogoh = \{x \in R : x^3 \in R - \{\pm 1\}\}$

$x \in R - \{\pm 1\}$

ii) $hofog$: $hof = \frac{1}{x^3}$ $\quad hofog = \frac{1}{(x^2-1)^3}$ $\quad x \in R - \{\pm 1\}$

iv) $fogof$: $fog = \frac{1}{x^2-1}$ $\quad fogof = \frac{1}{\frac{1}{x^2}-1}$ $\quad x \in R - \{\pm 1, 0\}$

iii) $fohog$: $\Rightarrow foh = \frac{1}{x^3}$ $\Rightarrow fohog = \frac{1}{(x^2-1)^3}$ $\quad x \in R - \{\pm 1\}$

4) β) $f_{2x} = \frac{1}{\sqrt{\ln(x^2+1)}}$, $g(x) = x^2+1$ $\Big\}$ $h(x) = \frac{1}{\sqrt{\ln(x)}}$ $\Big\}$ $\Rightarrow hog = \frac{1}{\sqrt{\ln(x^2+1)}}$

6γ) i) $f(x) = x-1$ (1-1 προ 1) ii) $f(x) = x^2-1$ (όχι 1-1)

7) α) $f(g(x)) = 4x+1$, $g(f(x)) = 4x-3$ $\Big\}$ Δεν είναι αντίστροφες

γ) $f(g(x)) = \frac{\frac{x-2}{x-1} - 1}{\frac{x-2}{x-1} - 2} = \frac{1}{x}$ $\quad g(f(x)) = \frac{\frac{x-1}{x-2} - 2}{\frac{x-1}{x-2} - 1} = -x+3$

$\Rightarrow$ Δεν είναι αντιστρ

8) α) $f(x) = 3x-4$

$y = \frac{1}{3}x + \frac{4}{3}$ $\quad f^{-1}(x) = \frac{1}{3}x + \frac{4}{3}$

β) Α

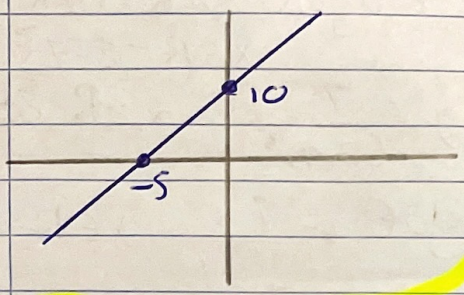
γ) $y = 3x^3+9 \Rightarrow f^{-1}(x) = \sqrt[3]{\frac{1}{3}x-3}$

δε) $y = \frac{x-1}{x-2} \Rightarrow f^{-1}(x) = \frac{-2x+1}{1-x}$

~~1.3)~~

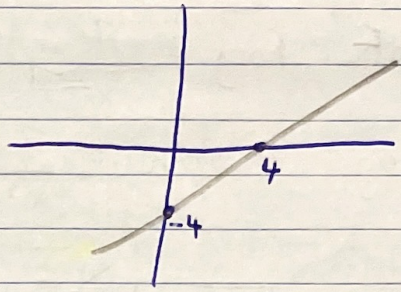
1.3) 1β) $x = t - 4$

$\boxed{t = x + 4} \Rightarrow y = 2(x+4) + 2 \Rightarrow \boxed{y = 2x + 10}$



1α) $x = t + 2$

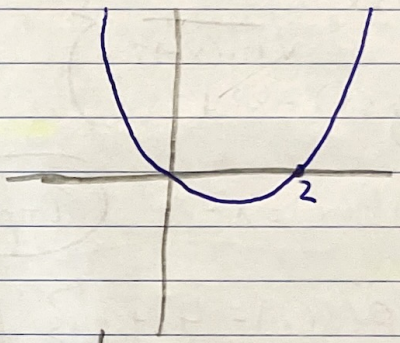
$\boxed{t = x - 2} \Rightarrow y = x - 4$



ε) $x = 2t + 1 \Rightarrow \boxed{t = \frac{x-1}{2}}$

$y = 4t^2 - 1$

$\Rightarrow y = 4\left(\frac{x-1}{2}\right)^2 - 1 = (x-1)^2 - 1$

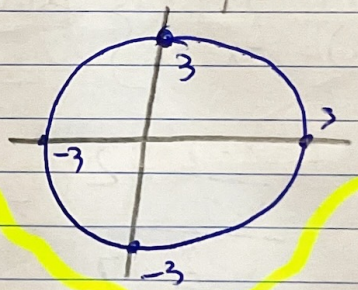


5) $x = 3\sin(\pi t) \quad y = 3\cos(\pi t)$

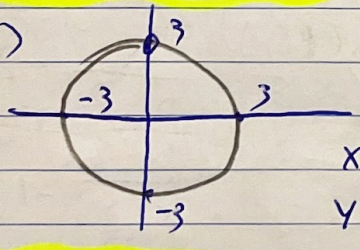
$x^2 + y^2 = 9\sin^2(\pi t) + 9\cos^2(\pi t)$

$= 9(\sin^2(\pi t) + \cos^2(\pi t)) = 9$

$\boxed{x^2 + y^2 = 9}$



2) a)



$x = 3\sin t$

$y = 3\cos t$

2.1) γ) $\lim_{x \to 0^+} g(x) = 3 \qquad \lim_{x \to -\infty} g(x) = \nexists \qquad g(0) = 3$

ε) $\lim_{x \to 0^-} F(x) = 7 \qquad \lim_{x \to 0^+} F(x) = \infty \qquad \lim_{x \to \alpha} F(x) = +\infty \qquad F(0) = -2$

στ) $\lim_{x \to 4} G(x) = 7 \qquad \lim_{x \to 0^-} G(x) = 3 \qquad \lim_{x \to +\infty} G(x) = 6$

$\lim_{x \to 0^+} G(x) = 3 \qquad \lim_{x \to -\infty} G(x) = \nexists$

①

2.2 $f(x)=3x \quad L=9 \quad \epsilon=0,1 \quad a=3 \quad 0<|x-1|<\delta$

$|3x-9|<0,1$

② a) $|-2x-7|<\epsilon \qquad 0<|x+3|<\delta$

$|-2x-6|<\epsilon$

$-2|x+3|<\epsilon \qquad |x+3|<\frac{\epsilon}{-2} \Rightarrow \lim_{x\to -3}(1-2x)=7$

5) a) $\left|\frac{2x}{x+1}-2\right|<\epsilon \qquad x>N \qquad 2x<(\epsilon-2)(x+1)$

$2x<\epsilon x+\epsilon-2x-2$

$\boxed{\lim_{x\to+\infty}\frac{2x}{x+1}=2} \Leftarrow \frac{2-\epsilon}{\epsilon}<x \quad N=\frac{2-\epsilon}{\epsilon}$

6a) $\lim_{x\to 1}\frac{2}{(x-1)^2}=\infty \qquad \frac{1}{(x-1)^2}>M \qquad (x-1)^2<\frac{1}{m}$

$-\frac{1}{\sqrt{m}}<x-1<\frac{1}{\sqrt{m}} \Rightarrow \delta=\frac{1}{\sqrt{m}}$

αρα $\Rightarrow \lim_{x\to 1}\frac{1}{(x-1)^2}=+\infty$

7) a) $\lim_{x\to a}[1+2]=3$ β) $\lim_{x\to a}[1+2.0]=1$ γ) $\lim_{x\to 2}\frac{0}{-2}=0$

δ) $\lim_{x\to a}\frac{2\cdot 1+(-2)}{0}=\frac{0}{0}$ (Απρ) ε) $\lim_{x\to a}[1(-2)]^2=4$

στ) $\lim_{x\to a}\sqrt{1-2}=\sqrt{-1}$ ∄

8) a) $\lim_{x\to 1^+}(x^3-2x^2+2x-1)=0$ δ) $\lim_{x\to\infty}(2x^2-3x^3+4x^4+x)$

β) $\lim_{x\to 1^-}(-2x^4-4x^3-2x^2)=0$ $=\lim_{x\to\infty}-3x^3=-\infty$

γ) $\lim_{x\to 3}[(x-2)(x-1)-(x+1)^2]=32$ στ) $\lim_{y\to -2}\sqrt{y^3+5}\Rightarrow$ (ΑΠΡ)

ε) $\lim_{x\to 1^+}\frac{1}{(x-1)^3}=+\infty$ ιβ) $\lim_{x\to -1}\frac{x^2+x}{x-1}=\frac{0}{-2}=0$

ιγ) $\lim_{x\to\infty}\frac{x^2-3x+2}{x-1}=\lim_{x\to+\infty}\frac{x^2}{x}=+\infty$

2.3 2β) $f(x)=\sqrt{7-x^2}$ $x^2 \geq 1$ $\boxed{x \geq 1}$

ε) $f(x)=\frac{x+1}{x^2+2x+3}$ ~~✗~~ αδυνατεί

στ) $f(x)=\frac{x+1}{x^2+3x+2}$ $x^2+3x+2 \neq 0$

$(x+2)(x+1) \neq 0$ $\boxed{x \neq -2}$ $\boxed{x \neq -1}$

1α) $f(x)=\begin{cases}\frac{x^2+1}{x-1}, & x>1 \Rightarrow x \neq 1 \\ x+1, & x \leq 1\end{cases}$

4) γ) $f(x)=\frac{x+1}{|x|+1}$ $\begin{cases}\frac{x+1}{x+1}=1, & x \geq 0 \\ \frac{x+1}{-x+1}, & x<0\end{cases}$

5) α) $x^3+x-5=0$, $[0,2]$

$f(0)=-5$, $f(2)=2$ $\Big\}$ $f(0)\cdot f(2)<0 \Rightarrow$ Bolzano $\exists$ τουλάχιστον μία ρίζα στο διάστημα $[0,2]$

2.4) 3) δ) $\lim\limits_{x \to 0} \frac{\sin x}{\cos x - 1} = \frac{0}{0}$ (Απρ) L'Hospital $\Rightarrow \lim\limits_{x \to 0} \frac{\cos x}{-\sin x} = \infty$

ν) $\lim\limits_{x \to 0} \frac{\tan 5x}{\sin 3x} = \frac{0}{0}$ Απρ (DLH) $5 \cdot \frac{\frac{1}{\cos^2 5x}}{3 \cdot \cos^3 x} = \frac{5}{3}$

5) β) $\lim\limits_{x \to \pi} \frac{x-\pi}{\sin 2x}$ (DLH) $\Rightarrow \lim\limits_{x \to \pi} \frac{1}{2\cos 2x} = \frac{1}{2}$

3.1) 1β) $\lim\limits_{x \to -2} \frac{f(x)-f(-2)}{x+2} = \lim\limits_{x \to -2} \frac{\frac{1}{x+1}+1}{x+2} = \lim\limits_{x \to -2} \frac{1+x+1}{(x+1)(x+2)}$

$= -\lim\limits_{x \to -2} \frac{x+2}{(x+1)(x+2)} = -1$

3.2) 1) β) $y' = \frac{1}{2\sqrt{x}} - \frac{3 \cdot 5}{2\sqrt{5x}} + \frac{3x^7}{2\sqrt{x^3}} + \frac{7}{4\sqrt{x}}$

γ) $y' = \frac{-1}{x^2} + \frac{3}{x^2} - \frac{6}{9x^2} + \frac{3}{9}$

ε) $y' = \frac{2x}{2\sqrt{x^2+5}}$

ν) $y' = 6x e^{3x^2}$

ι) $y' = 2x\sin x + x^2 \cos x$

ιδ) $y' = \sin\sqrt{x} + x \cdot \frac{1}{2\sqrt{x}} \cdot \cos\sqrt{x}$

ιγ) $y' = \frac{-2x\sin x - x^2 \cos x}{\sin^2 x}$

κοζ) $y' = \frac{(-\frac{3}{2}\sin 3x)(7-3x) + \frac{3}{2}\cos 3x}{(7-3x)^2}$

2) β) $\frac{2}{\frac{2\sqrt{2x+7}}{2}} = \frac{1}{2\sqrt{2x+7}}$ δ) $\frac{1}{\frac{2\sqrt{x}}{\sqrt{x}}} = \frac{1}{\sqrt{x}}$

3) α) $\frac{dv}{dx} = \frac{f'(x)}{2\sqrt{f(x)}}$ γ) $\frac{dv}{dx} = \frac{f'(x)}{f(x)}$ $\frac{1}{3\sqrt{x}}$

3.3) 7) δ) $y'' = \frac{x \cdot 2\sqrt{x^2+5} - 2x\frac{4x}{4\sqrt{x^2+5}}}{2\sqrt{x^2+5}} =$

ε) $y(5) = \frac{7}{2} \cdot e^x + 24e^{2x} + 162e^{-3x}$

3.4) (7) α) $2x + 3y + 3x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{4x - 3x^2y^3}{x^33y^2 - 4x}$

ε) $\sin y + x\cos y \cdot \frac{dy}{dx} + \frac{dy}{dx}\sin x + y\cos x = 7 + \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{1 - \sin y - y\cos x}{x\cos y + \sin y - 1}$

2α) $\frac{d^2y}{dx^2} = \frac{(-2 - 3\frac{dy}{dx})(3x+2y) - (-2x-3y)(3+2\frac{dy}{dx})}{(3x+2y)^2}$

3.5) (7) α) $\frac{dy}{dx} = 3t^2$ $\frac{dy}{dx} = 2t$ $\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2}{2t} = \frac{3}{2}t$

ος) $\frac{dy}{dx} = \frac{3\cos\theta}{-2\sin\theta}$ $\frac{\frac{dv}{d\theta}}{\frac{dy}{d\theta}} = \frac{3\cos\theta}{-2\sin\theta}$

3) $dy = 2x-4$

$dx = \frac{1}{2}(2t^2+7)^{3/2}\cdot(4t) = 2t(2t^2+7)^{3/2}$

$$\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2x-4}{2\sqrt{2}(2\cdot 2+7)^{3/2}}$$

4.7) $f(x)=x^2-4x+5$

x		2	
	−	0	+
	↘	min	↗

7a) $f'(x)=2x-4 \Rightarrow =0 \Rightarrow x=2$

Fθιν $(-\infty, 2]$

Fαυξ $[2, +\infty)$

3β) $y = x^{4/3}(7-x)^{7/3}$

$y' = \frac{4\sqrt[3]{x-x^2}}{3} - \frac{x^3\sqrt{x}}{3\cdot 3\sqrt[3]{(x)^2}} \Rightarrow y'=0$

θ) $y=\frac{x^2}{x^2+7}$ $y'=\frac{2x}{(x^2+7)^2}$ $y'=0 \Rightarrow \boxed{x=0}$

x		0	
	−	0	+
	↘	min	↗

φθιν στο $(-\infty, 0]$

αυξ στο $[0, +\infty)$

1β) $y=\cos^2 x$

$y' = -2\eta\mu x$ $y'=0 \Rightarrow \boxed{x=k\pi}$

4) $y' = 4x^3-4x$ $y''=12x^2-4$ $y''=0 \Rightarrow 12x^2=4$

$x^2=1/3$

$\boxed{x=\pm 1/\sqrt{3}}$

x		$-1/\sqrt{3}$		$1/\sqrt{3}$	
	+	0	−	0	+
	↗	max	↘	min	↗

n) $y'' = \frac{72}{(x-3)^3}$ $\boxed{x\neq 3}$

x		3	
	+	∄	+
	↗		↗

6) β) $f'(x)=6x^2+6x-12$

$f'(x)=0 \Rightarrow x^2+x-2=0$

$(x+2)(x-1)=0$

$\boxed{x=-2}$ $\boxed{x=1}$

x		−2		1	
	+	0	−	0	+
	↗	max	↘	min	↗

max(3, 45)

min(1, −7)

$f(-3)=9$ $f(3)=45$ $f(-2)=20$ $f(1)=-7$

ΚΑΡΥΟΣ.Γ

7) α) $f'(x) = -4 - 8x$

$f'(x) = 0 \Rightarrow 8x = -4$

$x = -\frac{1}{2}$



$f(-\frac{1}{2}) = 7 \quad (-\frac{1}{2}, 7)$

β) $f(x) = \frac{x+1}{x-1}$

$f'(x) = \frac{-2}{(x-1)^2}$ $\quad f'(x) \neq 0$

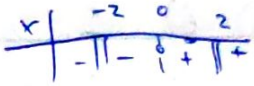


$f(-1) = 0 \quad (-1, 0)$

$f(1) = \nexists$

δ) $f(x) = \frac{2}{4-x^2}$

$f'(x) = \frac{4x}{(4-x^2)^2}$ $\quad x = 0, \; x = \pm 2$



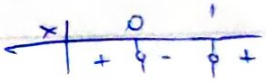
4.2

1) α) $f(x) = x^2 - 4x + 5$

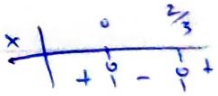
$f'(x) = 2x - 4$

$f''(x) = 2$

2) α) $f'(x) = 12x^3 - 12x^2 = 0 \Rightarrow x = 0, \; x = 1$

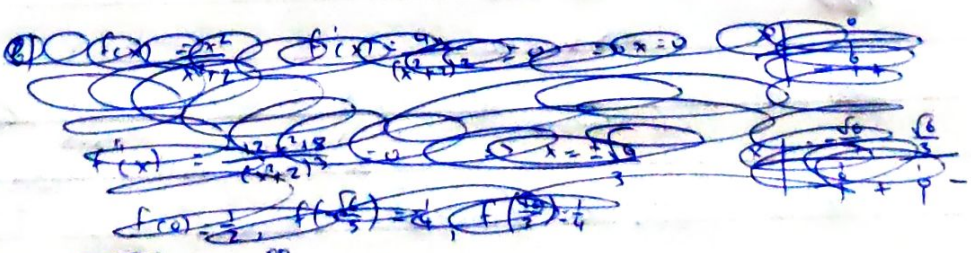


$f''(x) = 36x^2 - 24x = 0 \Rightarrow x = 0, \; x = \frac{2}{3}$



$f(0) = 0, \quad f(1) = -1, \quad f(\frac{2}{3}) = \frac{4}{27}$

4.5

1) 

ε) $\sum_{n=2}^{\infty} \frac{n^2}{2^n(n-1)}$

4.7

1) α) $f(x) = \cos x$

$f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$

2) B) $f(x) = \ln(2x-1)$

$f(x) = \frac{2(x-1)}{1!} + \frac{4(x-1)^2}{2!} + \frac{16(x-1)^3}{3!} \dots$

S1

1) B) $\int (7x^{\frac{3}{2}} - 5x^{\frac{5}{2}} + 3^{\frac{3}{2}} - x^{\frac{1}{3}})\,dx = \frac{14}{9}x^{\frac{9}{2}}\sqrt{x} - \frac{10}{7}x^{\frac{7}{2}}\sqrt{x} + \frac{6}{5}x^{\frac{5}{2}}\sqrt{x} - \frac{2\sqrt{x}}{3} + C$

S) $\int \frac{x^3 - x + 3}{2x^3}\,dx = \frac{x}{2} + \frac{1}{2x} - \frac{3}{4x^2} + C$

ζ) $\int \sqrt{3x-1}\,dx = \frac{2\sqrt{3x-1}\,(3x-1)}{9} + C$

i) $\int \frac{1}{(2-y)^2}\,dy = \frac{1}{2(2-y)^2} + C$

iot) $\int \frac{x+3}{1-2x}\,dx = -\frac{x}{2} - \frac{7}{4}\ln|1-2x| + C$

K) $\int \sec 3x \tan 3x\,dx = \frac{1}{3}\sec 3x + C$

S2

1) B) $I = \frac{4(t^3+t-2)^{\frac{7}{4}}}{7} + C$

S) $I = 3\int \frac{t}{(t^2+3)^{\frac{1}{3}}}\,dt = \frac{3}{2}\int 2t(t^2+3)^{-\frac{1}{3}}\,dt$

$= \frac{\frac{3}{2}(t^2+3)^{\frac{2}{3}}}{\frac{2}{3}} + C$

ζ) $I = \int \frac{x}{\sqrt{x^2+1}}\,dx = \frac{1}{2}\int 2x(x^2+1)^{-\frac{1}{2}}\,dx$

$= \frac{1}{2}\frac{(x^2+1)^{\frac{1}{2}}}{\frac{1}{2}} + C$

iot) $I = \int \frac{1}{\sqrt{12+4x-x^2}}\,dx = \int \frac{1}{\sqrt{-(x^2-4x+12)}}\,dx = \int \frac{1}{\sqrt{8-(x^2-4x+4-10)}}\,dx$

$= \int \frac{1}{\sqrt{[-(x-2)^2-16]}}\,dx$

$= \int \frac{1}{\sqrt{16-t^2}}\,dt$

$= \sin\left(\frac{t}{4}\right) = \sin\left(\frac{x-2}{4}\right) + C$

$t = x-2 \quad dt = dx$

$\Rightarrow x = t+2$

Lη) $I = \int \frac{x+1}{x^2-4x+8}\,dx = \int \frac{\frac{1}{2}(2x-4)}{x^2-4x+8} + \frac{3}{x^2-4x+8}\,dx$

$= \frac{1}{2}\ln(|x^2-4x+8|) + \frac{\arctan\left(\frac{x-2}{2}\right)}{2} + C$

K6) $I = \int \frac{\cos\sqrt{x}}{2\sqrt{x}}\,dx = \int \cos t\,dt = \sin t = \sin\sqrt{x} + C$

$*\Rightarrow t = \sqrt{x} \Rightarrow x = t^2$

$dt = \frac{1}{2\sqrt{x}}\,dx$

$dx = 2\sqrt{x}\,dt$

κε) $I = \int \sin^4 x \cos x \, dx$

$= \int \sin^4 x \cos x \, dx$ $\quad t = \sin x$

$= \int t^4 \, dt$ $\quad dt = \cos x \, dx$

$= \frac{t^5}{5} = \frac{\sin^5 x}{5} + c$ $\quad dx = \frac{dt}{\cos x}$

λα) $I = \int e^{\cos x} \sin x \, dx$ $\quad t = \cos x$

$= -\int e^t \, dt$ $\quad dt = -\sin x \, dt$

$= -e^{\cos x} + c$ $\quad dt = \frac{dt}{-\sin x}$

λβ) $I = \int (e^x + 1)^2 e^x \, dx$

$= \frac{(e^x+1)^3}{3} + c$

S.3

1) γ) $I = \int \frac{x}{x^2 - 3x - 4} \, dx = \frac{4}{3} \ln(x-4) + \frac{1}{5} \ln(x+1) + c$

$\Rightarrow \frac{x}{x^2-3x-4} = \frac{x}{(x-4)(x+1)} = \frac{A}{x-4} + \frac{B}{x+1} = A(x+1) + B(x-4)$

$= \frac{4}{5(x-4)} + \frac{1}{5(x+1)}$

$x = -1 \Rightarrow B = \frac{1}{3}$

$x = 4 \Rightarrow A = \frac{4}{5}$

η) $I = \int \frac{1+x}{(4+x^2)(x+2)} \, dx = \frac{\frac{x}{2} + \frac{3}{4}}{x^2+4} - \frac{1}{8(x+2)}$

$\frac{1+x}{(x^2+4)(x+2)} = \frac{Ax+C}{x^2+4} + \frac{B}{x+2} = (Ax+C)(x+2) + B(x^2+4) = \frac{\frac{x}{2} + \frac{3}{4}}{x^2+4} - \frac{1}{8(x+2)}$

$x = -2 \Rightarrow B = -\frac{1}{8}$

$x = 0 \Rightarrow C = \frac{3}{4}$

$x = 1 \Rightarrow A = \frac{1}{2}$

ιβ) $I = \int \frac{2x}{(x-1)^2(x+2)} \, dx = \frac{4}{9} \int \frac{1}{x-1} \, dx + \frac{2}{3} \int (x-1)^{-2} \, dx = \frac{4}{9} \int (x-1)^2 \, dx = \frac{4}{9} \ln(x-1) +$

$\Rightarrow \frac{2}{3} \frac{(x-1)^{-1}}{-1} - \frac{4}{9} \frac{(x-1)^{-1}}{-1} + C$

$\frac{2x}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$.

$x = -2 \Rightarrow C = -\frac{4}{9}$

$x = 1 \Rightarrow B = \frac{2}{3}$

$x = -1 \Rightarrow A = \frac{4}{9}$

5.4

i) b) $I = \int x^2 e^x dx = x^2 e^x - 2\int e^x x\,dx$

$= x^2 e^x - 2[xe^x - \int e^x dx]$

$= x^2 e^x - 2xe^x + 2e^x + c$

ε) $I = \int 2x\cos 2x\,dx = x\sin 2x - \int \sin 2x\,dx$

$= x\sin 2x + \frac{\cos 2x}{2} + c$

ι) $I = \int x^2 \ln x\,dx = \int \ln x\, d\left(\frac{x^3}{3}\right) = \frac{x^3}{3}\ln x - \int \frac{x^2}{3}dx = \frac{x^3}{3}\ln x - \frac{1}{3}x^3 dx$

$= \frac{x^3}{3}\ln x - \frac{x^3}{9} + c$

ν) $I = \int e^{2x}\sin x\,dx = \sin x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2}\cos x\,dx = \frac{1}{2}\left(\cos x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2}\right)$

γ) $I = \int \tan^{-1} 2x\,dx = \int \frac{\tan^{-1} t}{2}\,dt$

(1) $t = 2x$

$dt = 2dx$

$dx = \frac{dt}{2}$

(2) $u = 1+t^2$

$du = 2t\,dt$

$= \frac{1}{2}\int \tan^{-1} t\,dt = \frac{1}{2}\left[\tan^{-1} t\cdot t - \frac{1}{1+t^2}dt\right]$

$= \frac{1}{2}\left[t\tan^{-1} t\cdot t - \int \frac{1}{2u}du\right]$

$= \frac{1}{2}\left[t\tan^{-1} t\cdot t - \frac{1}{2}\ln|u|\right]$

$= x\tan^{-1} 2x - \frac{1}{4}\ln|1+4x^2| + c$

5.6

i) b) $\int \frac{1}{x^2\sqrt{16-x^2}}dx = \int \frac{\cos u}{4\sin^2 u\sqrt{16-16\sin^2 u}}du = \int \frac{\cos u}{4\sin^2 u\sqrt{16\cos^2 u}}du$

$x = 4\sin u$

$\sin u = \arcsin \frac{x}{4}\,dx$

$dx = 4\cos u\,du$

$= \frac{1}{16}\int \frac{1}{\sin^2 u}du = \frac{1}{16}\int \cos^2 u\,du$

$= -\frac{\cos u}{16} = \frac{\sqrt{16-x^2}}{16x}$

ε) $I = \int \frac{2x^2}{\sqrt{9+x^2}}dx = \int \frac{x^2}{\sqrt{9+x^2}}dx = \int \frac{27\sec^2 u\tan^2 u}{\sqrt{9+9\tan^2 u+9}}$

$x = 3\tan u$

$u = \arctan\left(\frac{x}{3}\right)$

$dx = 3\sec^2 u\,du$

$= 9\int \sec u\tan^2 u\,du = 9\int \sec u(\sec^2 u - 1)du$

$= 9\int \sec^3 u - \sec u)du = \int \frac{\sec u\tan u}{2} + \frac{1}{2}\sec du$

$= \frac{\ln|\tan u + \sec u|}{2} + \frac{\sec u\tan u}{2}$

6.1

1) b) $I = \int_1^2 x(1+x^3)\,dx = \int_1^2 (x+x^4)\,dx = \left[\frac{x^2}{2} + \frac{x^5}{5}\right]_1^2 = \frac{81}{10}$ cm

α) $I = \int_{-1}^{1} x^3(x^4+1)^{-3}\,dx = -\frac{1}{8}\left[(x^4+1)^{-2}\right]_{-1}^{1} = 0$ cm

β) $I = \int_1^2 (x^2+1)\ln x\,dx = \int_1^2 (x^2\ln x + \ln x)\,dx$

$= \int_1^2 x^2\ln x\,dx + \int_1^2 \ln x\,dx$

$= \frac{x^3}{3}\ln x - \int_1^2 \frac{x^3}{3}\cdot\frac{1}{x}\,dx + x\ln x - \int_1^2 x\cdot\frac{1}{x}\,dx$

$= \left[\frac{x^3}{3}\ln x - \frac{x^3}{9} + x\ln x - x\right]_1^2$

$= \frac{11}{3}\ln 2 - \frac{16}{9}$

6.2

1) a) $\int_1^6 (9x - x^2)\,dx = \left[\frac{9x^2}{2} - \frac{x^3}{3}\right]_1^6 = \frac{516}{6}$ cm

δ) $\int_0^{\pi} x\cos\frac{x}{2}\,dx = \int_0^{\pi} x\,d\left(2\sin\frac{x}{2}\right) = 2x\sin\frac{x}{2} - 2\int_0^{\pi}\sin\frac{x}{2}\,dx$

$= \left[2x\sin\frac{x}{2} + 4\cos\frac{x}{2}\right]_0^{\pi}$

$= (2\pi - 4)$ cm

2) b) $\int_0^2 (4x - x^2)\,dx = \left[\frac{4x^2}{2} - \frac{x^3}{3}\right]_0^2 = \frac{16}{3}$ cm

6.3

3) a) $\int_0^{1/100} 400\sin\pi t\,dt = 400\int_0^{1/100}\sin(\pi t)\,dt = \frac{-400\cos\left(\frac{\pi}{100}\right) + 400}{\pi/100}$

b) R.M.S $\quad 160000\int_0^{1/100}\sin^2\pi t\,dt = \frac{800\pi - 4000\sin\left(\frac{\pi}{50}\right)}{100\pi}$

$RMS = \sqrt{\frac{800\pi - 4000\sin\pi/50}{100\pi}} = \sqrt{0,52} = 0,72$

7) $MV = \frac{\frac{\pi^3}{2} + \frac{2\pi(-1)}{2}}{5\pi} = \frac{\pi^2 - 2}{10}$

6.4

1) a) $V = \pi\int_0^1 \left[0^2 - (x+1)^2 \cdot x^4\right]dx = \pi\int_0^1 (x^2 + 2x + 1 - x^4)\,dx = \frac{32\pi}{15}$ cm

β) $V = \pi\int_{-2}^{2} (4 - x^2)^2\,dx = \frac{512\pi}{15}$ cm

3) γ) $V = \pi\int_0^4 \left[\left(\frac{x}{3}\right)^2 - (4-x)^2\right]dx = -\frac{512}{27}$ cm